# Power In Waves

> SYNOPSIS
>
> We round up the notes with equations for calculating power in waves.

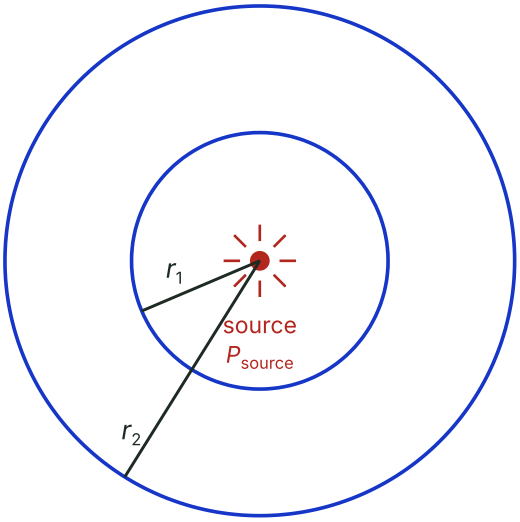
> BEFORE YOU READ ON — PREDICT
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> You are standing in an open field listening to a loudspeaker, and you walk until you are **twice as far away**. **(1)** What happens to the *intensity*? **(2)** What happens to the *amplitude* of the wave? **(3)** By how many *decibels* does the level drop? Write down your three answers. This note gives you all three, and the third one is a number worth carrying around.

## 3. Power in Waves

Note 4a counted the energy an oscillator carries. This note follows that energy outwards: how fast it leaves the source, how thinly it is spread by the time it reaches you, and why we end up measuring it on a logarithmic scale.

1. Write **power** and **intensity** as definitions, and give the SI unit of each.
2. A source radiates $P_{\text{source}}$ equally in every direction. At a distance $r$, what area is that power spread over? Write $I(r)$.
3. Write $I_1$ and $I_2$ at two distances and form the ratio $I_1/I_2$. What cancels, and what does that tell you?
4. Intensity turns out to go as the **square of the amplitude**. Given $E = \frac{1}{2}kA^2$ for a sinusoidal wave, why is that the relationship you should have expected?

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Label the intensity on each sphere and write it in terms of $P_{source}$ and the radius. What is the area of a sphere, and what cancels when you divide one intensity by the other?

**THE PUNCHLINE**

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The **inverse-square law**, and notice what it is *not*: nothing here is being used up. The source puts out the same power whether you stand near it or far from it; all that happens is that the power gets spread over a bigger and bigger sphere. Every inverse-square law in physics — sound, light, gravity, the electric field — is this same piece of geometry, which is why the exponent is exactly 2 and not 1.9.

## Sound intensity level (decibel)

A quiet room and a jet engine differ in intensity by about a factor of a trillion. No linear scale is going to show both ends of that comfortably, so sound is measured logarithmically, against a fixed reference.

5. Write the definition of the sound intensity level in decibels, and say what the reference $I_0$ is.

6. What does an increase of **10 dB** mean, in terms of intensity? And what level does a sound at the reference intensity itself have?

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## Looking ahead

Everything so far has had the source and the listener standing still. **Note 5** sets them moving: when a source or an observer is travelling, the frequency that arrives is not the frequency that left.

**Where this sits.** Note 4a found the energy stored in one oscillator and it went as $A^2$. This note sends that energy out into space, and the $A^2$ survives the trip: intensity goes as amplitude squared at every distance. The $1/r^2$ is not physics on top of that — it is the geometry of a sphere, and it would be true of anything spreading evenly in three dimensions.

BACK TO YOUR PREDICTION

**(1) The intensity drops to a quarter.** $I \propto 1/r^2$, so doubling $r$ divides $I$ by 4 — while the *power* of the loudspeaker has not changed at all. **(2) The amplitude halves.** $I \propto A^2$, so a factor of 4 in intensity is a factor of 2 in amplitude. **(3) Down 6 dB** — (10 dB) $\log_{10}(¼) = -6.02$ dB. **Every doubling of distance costs you 6 dB**, whatever the source and whatever the starting level, which is the one number from this note worth remembering by itself.

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